

Learning Objectives

1. Compare and contrast the relationships between lung volumes and capacities.
2. Discuss spirometry and its limitations in measurements of lung volumes and capacities.
3. Outline methods to determine residual volume and functional residual capacity.
4. Extrapolate lung volumes and capacities with respect to obstructive and restrictive lung diseases.
5. Demarcate regions of the lung that do not participate in gas exchange and learn methods to measure dead space.

Basic Concepts in Respiratory Physiology

1. Lung Volumes and Capacities
2. Pulmonary Test : Spirometry
3. Lungs Conditions: Restrictive and
Obstructive Disease
4. Dead Space

Lung Capacities

IC: $VT + IRV$

The maximum amount of air that can be inspired following a normal expiration

VC: $VT + IRV + ERV$ or Forced Vital Capacity (FVC)

The maximum amount of air that can be expired following a maximal Inspiration

Most you can exhale

FRC: $RV + ERV$

The amount of air remaining in the lungs following a normal expiration.

Volume when the system is relaxed.

Chest wall pulling out = lungs pulling in

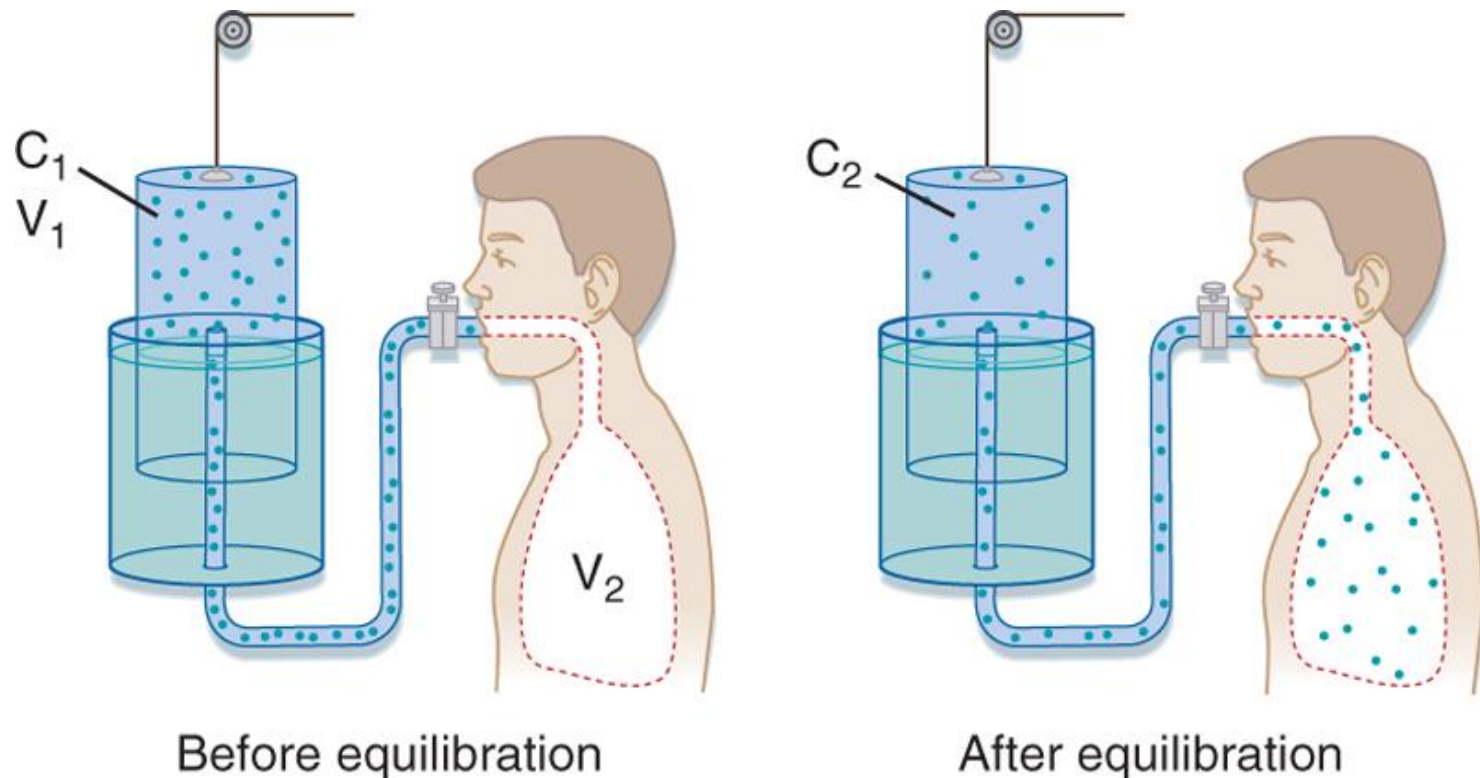
TLC: $RV + ERV + IRV + VT$ Sum of all volumes

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Helium Dilution Technique

An indirect method to measure the residual volume. Essential to know FRC which is important parameter for pulmonary diseases



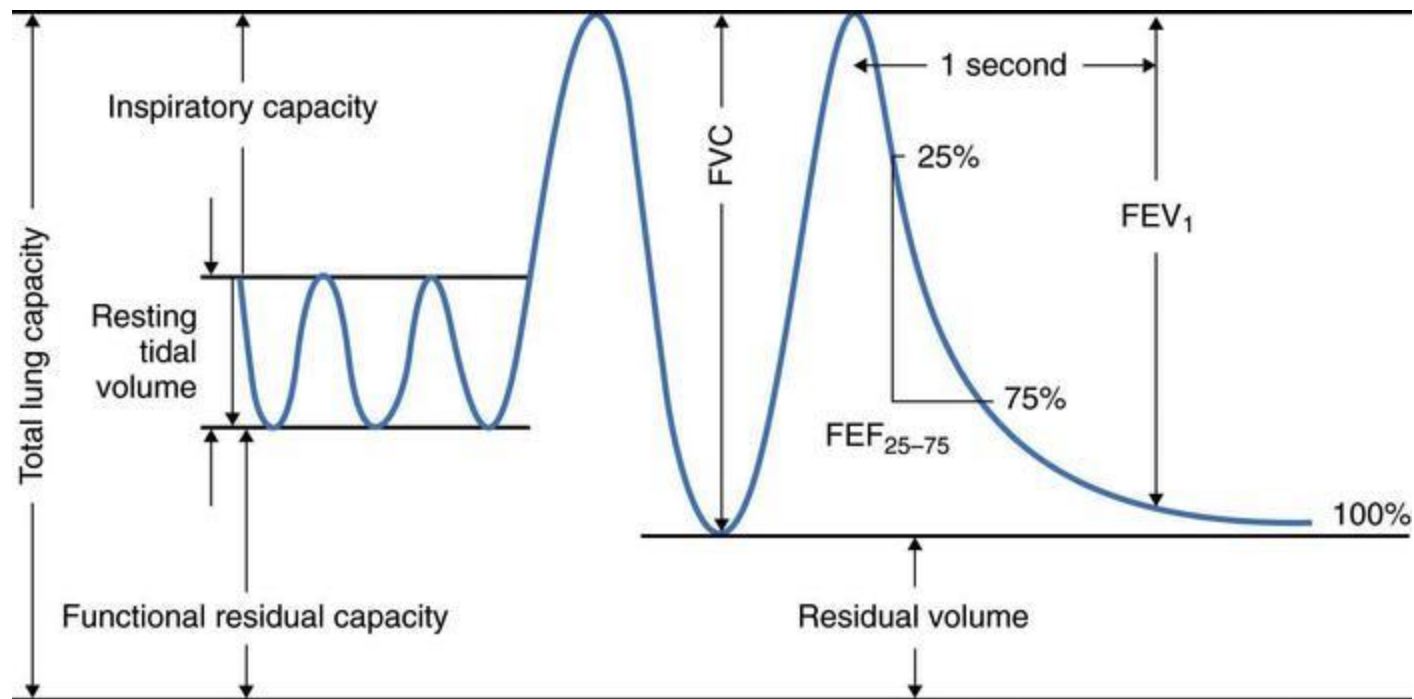
$$C_1 \times V_1 = C_2 \times (V_1 + V_2)$$

Koeppen & Stanton: Berne and Levy Physiology, 6th Edition.

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Forced Vital Capacity (FVC)

- Air exhaled as rapidly as possible after a maximum inhalation



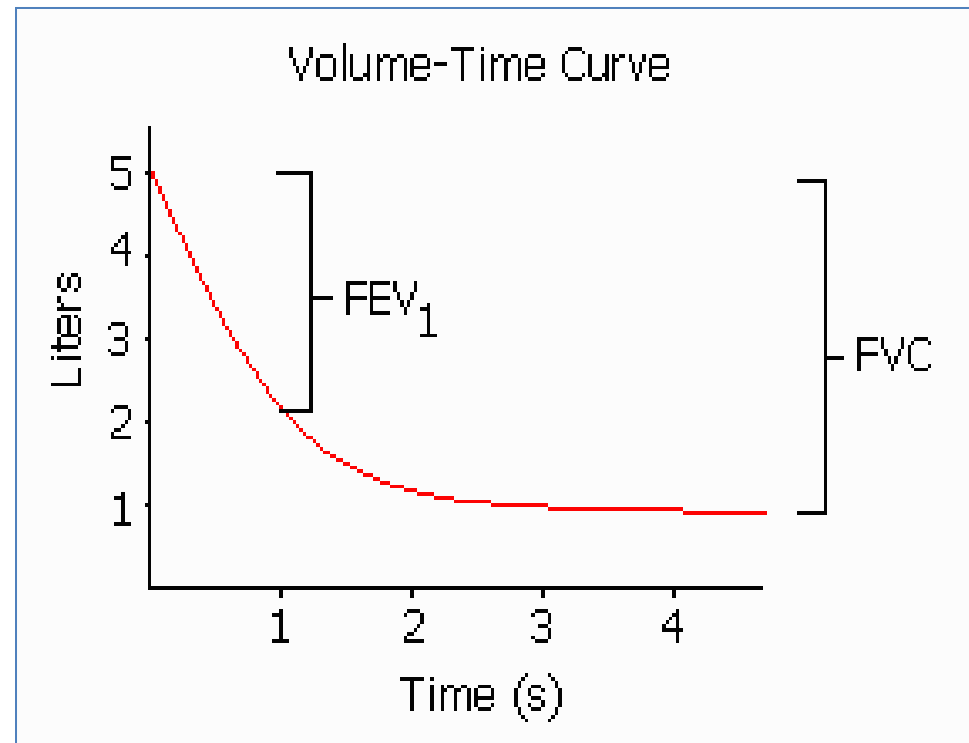
Brief Summary

FVC (Forced Vital Capacity) -- This is the maximum volume of gas that can be exhaled as forcefully and rapidly as possible after a maximal inhalation.

FEV₁ (Forced Expiratory Volume in 1st Second) -- This is the volume of air expired in the first second during maximal expiratory effort.

FEV₁/FVC -- This is the percentage of the vital capacity which is expired in the first second of maximal expiration. In healthy patients the FEV₁/FVC is usually around 70%.

Values are reported both as absolute values and as a percentage



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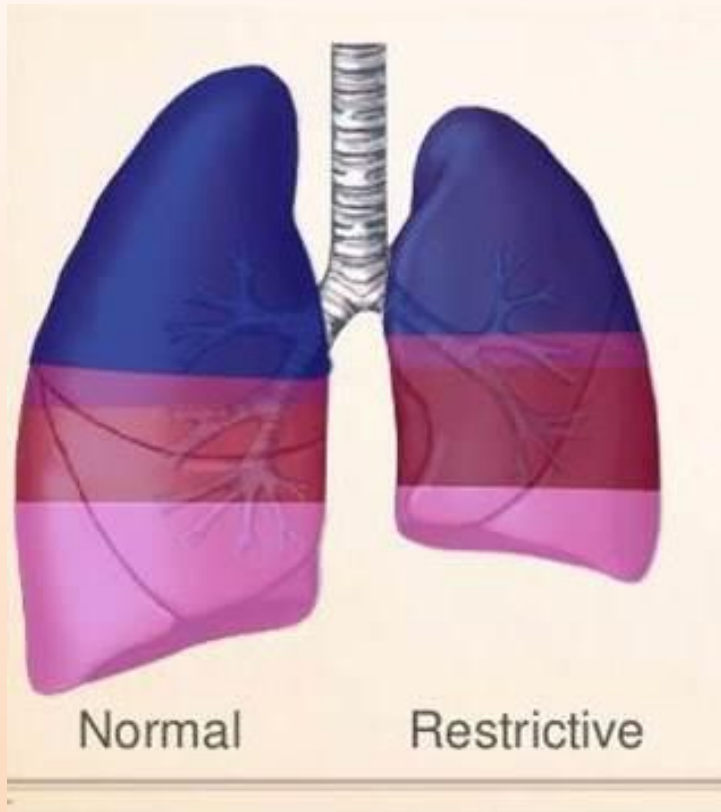
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Obstructive Lung Disease



- Airway obstruction causes an *increase in resistance*. **Air trapping**
- Common obstructive diseases include asthma, bronchitis, and emphysema.
- $\downarrow$ FEV1, because of $\uparrow$ airway resistance
- FEV1/FVC decreased
- This is because it is very difficult for a person with an obstructive disease (eg. asthma) to exhale quickly due to the increase in airway resistance.
- RV $\uparrow$

Restrictive Lung Disease



- The compliance of the lung is reduced, which increases the stiffness of the lung and limits expansion.
- Common causes of decreased lung compliance are pulmonary fibrosis, pneumonia and pulmonary edema.
- ↓ FVC.
- FEV1 is relatively large in comparison i.e. the FEV1/FVC ratio can be higher than normal.
- This is because it is easy for a person with a restricted lung to breathe out quickly, because of the high elastic recoil of the stiff lungs
- RV ↓

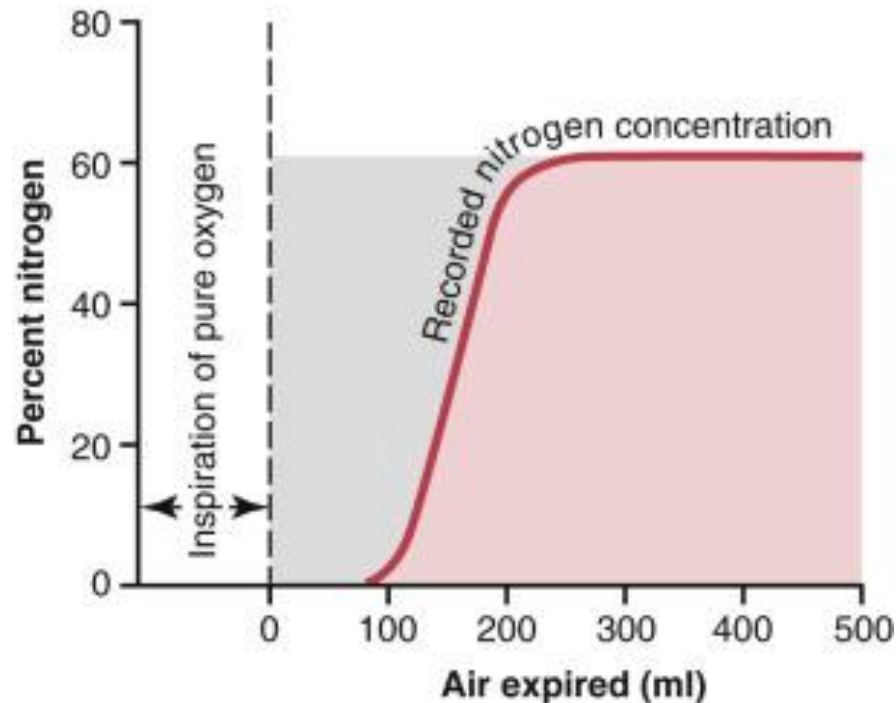


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Measurement of Anatomical Dead Space

Nitrogen Dilution - Fowler's Method



$$V_D = \frac{\text{Gray area} \times V_E}{(\text{Pink area} + \text{Gray area})}$$

where V_D is dead space air
and V_E is the total volume of
expired air.

A record of the changes in nitrogen concentration in the expired air after a single previous inspiration of pure oxygen.

Measurement of Anatomical and Alveolar Dead Spaces

TV multiplied by the dilution of alveolar PCO_2 by dead space air, which does not participate in gas exchange and does not therefore contribute CO_2 to expired air.

Bohr-Enghoff Method

$$\frac{V_D}{V_T} = \frac{P_{ACO_2} - P_{ECO_2}}{P_{ACO_2}}$$

V_D = physiologic dead space (mL)

V_T = tidal volume (mL)

P_{ACO_2} = PCO_2 of alveolar gas (mm Hg) = P_{aCO_2} of arterial blood

P_{ECO_2} = PCO_2 of expired air (mm Hg)

Summary

Most of the lung volumes and capacities can be directly measured via spirometry.

The residual volume (RV) can serve a valuable indicator of pulmonary disease and is derived indirectly.

Restrictive and obstructive lung disease can be distinguished by duration and volume of expiratory air release.

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